

Smart City Bike Sharing Data Analysis

Overview:

The dataset covers 404 bike stations across multiple cities, with a total of 21K available bike stands and 17K available bikes, reflecting a Bike Availability Rate of 22.35% — indicating low availability across the network.

Station Performance :

Station Occupancy Rate stands at 51.94%, meaning only half the bike stand capacity is being utilized. Out of all recorded entries, 70K are active stations while 4K entries show empty stations with zero bikes available — highlighting a significant supply gap.

Monthly Trend :

Bike availability peaked in August and November, then sharply declined through the following months — suggesting seasonal demand patterns or maintenance issues in off-peak months.

Operational Insight:

Almost 100% of recorded stations are OPEN status, confirming high operational uptime. However, the low bike availability rate suggests bikes are either in use, not returned, or not replenished at stations.

City Level Insights:

Cities with higher bike stands (20–40 stands per station) show better occupancy rates ranging.

Top 3 city station has high bike availability and as well as the available bike stands also.

- 00007 -Place victor Hugo
- 00030 – Place Anatole France
- 00087 – Lascrosses Lecerlc
- The station occupancy rate is high in “Q4” When compared to the other quarter in year 2024.
- Active station is 418 on each last update.

- Many station are in open in under Marseille contractor.